DATA 427 Group Project: Spark ML on the KDD Dataset

Insert ur names here

1 Implement two programs that apply Spark's Decision Tree algorithm and Logistic Regression algorithm to the provided KDD dataset. Run these programs 10 times on the school's Hadoop cluster, each using a different seed to split the dataset into a training and a test set.

1.1 Describe the two programs using pseudo-code

1.1.1 Logistic Regression:

Algorithm 1 Spark Logistic Regression

```
Parse the input/output directories and number of runs
Initialise the Spark session and load the data from the provided input directory

Setup the pipeline as follows
Index required columns
One-hot encode required columns
Assemble into feature columns
Append parameterised logistic regression model as the final stage.
for Number of runs do
    Set seed
    The data into train/test
    Fit the LR model on the train data
    Evaluate the models performance on the training and testing split, and record results
end for
Display results
```

1.2 Describe how to install and run the two programs step by step in a Readme.txt file.

Paste readme file here I guess?

1.3 Report the training and test results including the max, min, average accuracy and the standard deviation obtained from the 10 runs, and the running time of each program.

1.3.1 Technically these are local results, just putting here to get the format. (Also they will be pretty much the same on the cluster) Shoutout to generative ai for formatting my tables.

Table 1: Summary of Spark logistic regression accuracy over 10 runs.

Metric	Average	Std. dev.	Min	Max
Training accuracy (%)	94.35	0.12	94.15	94.49
Test accuracy (%)	94.24	0.15	94.01	94.45

Table 2: Spark logistic regression results for each seeded run.

Run	Seed	Train count	Test count	Train acc. (%)	Test acc. (%)
1	1	33506	14230	94.35	94.30
2	2	33450	14286	94.48	94.01
3	3	33508	14228	94.37	94.41
4	4	33466	14270	94.17	94.34
5	5	33376	14360	94.32	94.03
6	6	33581	14155	94.49	94.28
7	7	33458	14278	94.43	94.26
8	8	33561	14175	94.27	94.23
9	9	33435	14301	94.15	94.45
10	10	33353	14383	94.43	94.08

Table 3: Elapsed time for Spark logistic regression repeated executions.

Run	Seed	Time 1 (s)	Time 2 (s)
1	1	28.27	24.30
2	2	20.88	14.84
3	3	26.32	20.60
4	4	24.55	19.67
5	5	19.59	17.59
6	6	23.29	23.23
7	7	23.93	16.30
8	8	18.39	15.94
9	9	22.99	21.39
10	10	23.36	18.07
Average		23.16	19.19
Std. dev.		2.97	3.18
Min		18.39	14.84
Max		28.27	24.30